A Non-Hilbert UMD Subcase of General Characteristic Domination

Agent agent_lane_19

2026-06-16

Status

This packet gives a candidate partial result, likely valid, for Remark 12.10 of Yaroslavtsev's paper *Local characteristics and tangency of vector-valued martingales* [1]. It proves the general-local-martingale characteristic-domination estimate for the non-Hilbert UMD spaces ℓ_q, $1 < q < \infty$, at the natural exponent $p = q$.

The result is partial relative to the source question because it does not cover all exponents p in ℓ_q, all UMD spaces, or nonatomic L^q-spaces. It is nevertheless a genuine non-Hilbert positive case when $q \neq 2$.

Source Problem

The source is arXiv:1907.11588, PDF page 105, Remark 12.10. The source crop below states that the general-local-martingale case is open and reduces the missing accessible-jump issue to a discrete independent-increment comparison.

(12.11), (12.14), (2.9), and the fact that M and N are quasi-left continuous. $\square$

Remark 12.10. It is not known whether Theorem 12.8 holds for general local martingales. By Theorem 3.39 and by Proposition B.1 the main issue here is in proving a similar statement for discrete martingales with independent increments. Let us state this problem here as open. Let X be a Banach space, $1 \leq p < \infty$. Let $(\xi_n)_{n\geq 1}$ and $(\eta_n)_{n\geq 1}$ be X-valued mean-zero independent random variables such that for any Borel $B \in X \setminus \{0\}$

$$\sum_n \mathbb{P}(\xi_n \in B) \leq \sum_n \mathbb{P}(\eta_n \in B).$$

Does there exists a constant C (perhaps depending on p and X) such that

$$\mathbb{E}\Big\|\sum_n \xi_n\Big\|^p \leq C\mathbb{E}\Big\|\sum_n \eta_n\Big\|^p?$$

The next page adds that, by symmetrization, one may assume the variables are symmetric, and that even this symmetric discrete case was not known to the author.

By a standard symmetrization trick [66, Lemma 6.3] one can assume that ξ_n's and η_n's are symmetric. But even the symmetric case is not known for the author.

Main Result

Theorem 1. *Let $1 < q < \infty$. Let M and N be ℓ_q-valued local martingales such that N is characteristically dominated by M in the sense of [1]. Then*

$$\mathbb{E} \sup_{t\geq 0} \|N_t\|_{\ell_q}^q \leq C_q \, \mathbb{E} \sup_{t\geq 0} \|M_t\|_{\ell_q}^q ,$$

whenever the right hand side is finite.

Idea of the Proof

The Hilbert proof cannot simply be copied into ℓ_q, since $\|x\|^2$ is no longer generated by an inner product. The replacement is to use the lattice square-function theorem in exactly the exponent where the ℓ_q-norm is additive across coordinates.

Write $M^k = \langle M, e_k^* \rangle$ and $N^k = \langle N, e_k^* \rangle$. Characteristic domination pushes through the coordinate projection e_k^*, so each scalar martingale N^k is characteristically dominated by M^k. The scalar case is the one-dimensional Hilbert case, already supplied by the companion Hilbert all-p packet. Hence scalar BDG gives

$$\mathbb{E}[N^k]_\infty^{q/2} \lesssim_q \mathbb{E}[M^k]_\infty^{q/2}.$$

Now apply Yaroslavtsev's UMD Banach-function-space BDG theorem [2] to $\ell_q = L^q(\mathbb{N})$:

$$\mathbb{E} \sup_t \|N_t\|_{\ell_q}^q \simeq_q \mathbb{E} \left\| \left([N^k]_\infty^{1/2} \right)_{k\geq 1} \right\|_{\ell_q}^q = \sum_{k\geq 1} \mathbb{E}[N^k]_\infty^{q/2}.$$

Summing the scalar estimates and applying the same square-function theorem to M gives the result.

Two Inputs

Lemma 1 (Scalar characteristic domination)**.** *Let A and B be real-valued local martingales with $A_0 = B_0 = 0$. If A is characteristically dominated by B, then for every $1 < q < \infty$,*

$$\mathbb{E}[A]_\infty^{q/2} \leq C_q \mathbb{E}[B]_\infty^{q/2}.$$

Proof. The companion Hilbert packet [4] proves the characteristic-domination estimate for Hilbert-valued martingales for every finite exponent. Applied to the one-dimensional Hilbert space $\mathbb{R}$, it gives

$$\mathbb{E} \sup_{t\geq 0} |A_t|^q \leq C_q \mathbb{E} \sup_{t\geq 0} |B_t|^q.$$

The scalar Burkholder-Davis-Gundy inequalities convert both maximal moments to quadratic-variation moments:

$$\mathbb{E}[A]_\infty^{q/2} \simeq_q \mathbb{E} \sup_{t\geq 0} |A_t|^q, \qquad \mathbb{E}[B]_\infty^{q/2} \simeq_q \mathbb{E} \sup_{t\geq 0} |B_t|^q.$$

□

Lemma 2 (ℓ_q square function for local martingales)**.** *Let $1 < q < \infty$, and let L be an ℓ_q-valued local martingale with $L_0 = 0$. Write $L^k = \langle L, e_k^* \rangle$. Then*

$$\mathbb{E} \sup_{t \geq 0} \|L_t\|_{\ell_q}^q \simeq_q \sum_{k \geq 1} \mathbb{E}[L^k]_\infty^{q/2}.$$

Proof. The space ℓ_q is the UMD Banach function space $L^q(\mathbb{N}, \#)$. Yaroslavtsev's Banach-function-space BDG theorem [2, Theorem 5.9] gives, for the coordinate martingale field L^k,

$$\mathbb{E} \sup_{t \geq 0} \|L_t\|_{\ell_q}^q \simeq_q \mathbb{E} \left\| \left([L^k]_\infty^{1/2} \right)_{k \geq 1} \right\|_{\ell_q}^q.$$

The last expression is

$$\mathbb{E} \sum_{k \geq 1} [L^k]_\infty^{q/2} = \sum_{k \geq 1} \mathbb{E}[L^k]_\infty^{q/2}$$

by Tonelli's theorem. □

Proof of the Main Result

First assume $M_0 = N_0 = 0$. For each coordinate k, let $\pi_k : \ell_q \to \mathbb{R}$ be the bounded coordinate projection. We claim that $\pi_k N = N^k$ is characteristically dominated by $\pi_k M = M^k$.

The continuous part follows directly from the bilinear quadratic-variation order in the definition of characteristic domination:

$$[N^{k,c}]_\infty = [\langle N^c, e_k^* \rangle]_\infty \leq [\langle M^c, e_k^* \rangle]_\infty = [M^{k,c}]_\infty.$$

For the jump part, if $A \subset \mathbb{R} \setminus \{0\}$ is Borel, the jump compensator of the scalar coordinate is the pushforward of the vector jump compensator:

$$\nu^{N^k}(\mathbb{R}_+ \times A) = \nu^N(\mathbb{R}_+ \times \pi_k^{-1}(A)).$$

Since $\pi_k^{-1}(A) \subset \ell_q \setminus \{0\}$, characteristic domination of N by M gives

$$\nu^{N^k}(\mathbb{R}_+ \times A) \leq \nu^M(\mathbb{R}_+ \times \pi_k^{-1}(A)) = \nu^{M^k}(\mathbb{R}_+ \times A).$$

The initial condition is zero in the reduced case. Thus N^k is characteristically dominated by M^k.

Lemma 1 yields

$$\mathbb{E}[N^k]_\infty^{q/2} \leq C_q \, \mathbb{E}[M^k]_\infty^{q/2} \qquad (k \geq 1).$$

Summing over k and applying Lemma 2 to N and M,

$$\begin{aligned} \mathbb{E} \sup_{t \geq 0} \|N_t\|_{\ell_q}^q &\lesssim_q \sum_{k \geq 1} \mathbb{E}[N^k]_\infty^{q/2} \\ &\leq C_q \sum_{k \geq 1} \mathbb{E}[M^k]_\infty^{q/2} \\ &\lesssim_q \mathbb{E} \sup_{t \geq 0} \|M_t\|_{\ell_q}^q. \end{aligned}$$

For general initial values, set $\widetilde{M}_t = M_t - M_0$ and $\widetilde{N}_t = N_t - N_0$. The continuous and jump characteristics are unchanged, so the zero-initial estimate applies to $\widetilde{N}, \widetilde{M}$. The initial part is controlled coordinatewise: the definition of characteristic domination gives $|N_0^k| \leq |M_0^k|$ for every k, hence

$$\mathbb{E} \|N_0\|_{\ell_q}^q = \sum_k \mathbb{E}|N_0^k|^q \leq \sum_k \mathbb{E}|M_0^k|^q = \mathbb{E} \|M_0\|_{\ell_q}^q \leq \mathbb{E} \sup_t \|M_t\|_{\ell_q}^q.$$

Using $\sup_t \|N_t\| \leq \|N_0\| + \sup_t \left\| \widetilde{N}_t \right\|$ and the analogous bound for $\widetilde{M}$ proves Theorem 1.

Discrete Consequence

Corollary 1. *Let $1 < q < \infty$. Let $(\xi_n)_{n=1}^N$ and $(\eta_n)_{n=1}^N$ be independent mean-zero ℓ_q-valued random variables such that*

$$\sum_{n=1}^N \mathbb{P}(\xi_n \in B) \le \sum_{n=1}^N \mathbb{P}(\eta_n \in B) \qquad (B \subset \ell_q \setminus \{0\} \text{ Borel}).$$

Then

$$\mathbb{E} \max_{1\le m\le N} \left\| \sum_{n=1}^m \xi_n \right\|_{\ell_q}^q \le C_q \mathbb{E} \max_{1\le m\le N} \left\| \sum_{n=1}^m \eta_n \right\|_{\ell_q}^q .$$

In particular the same estimate holds for the terminal sums.

Proof. Put the increments at deterministic times $1, \ldots, N$. Independence makes the predictable jump compensator of the ξ-sum equal to $\sum_n \mathcal{L}(\xi_n)$, and similarly for the η-sum. The aggregate domination hypothesis is exactly characteristic domination of these discrete-time martingales; the continuous characteristics and initial values are zero. Theorem 1 gives the maximal estimate, and the terminal estimate follows by dropping the maximum. □

Limitations and Novelty Check

This packet does not settle the source problem for arbitrary p in ℓ_q. The proof uses the identity

$$\left\| \left(a_k^{1/2}\right)_k \right\|_{\ell_q}^q = \sum_k a_k^{q/2},$$

so the exponent $p = q$ is built into the summation step. A different tail or good-lambda argument would be needed to decouple the martingale moment exponent from the lattice exponent.

The proof also does not directly cover nonatomic $L^q(S)$. The coordinate pushforward argument relies on bounded coordinate functionals and on $\ell_q = L^q(\mathbb{N})$ being atomic. Point evaluations are not bounded linear functionals on a general $L^q(S)$, so the same proof has no immediate nonatomic formulation.

Cheap run indexes contained the previous c_0 counterexample and Hilbert packets, but no non-Hilbert ℓ_q packet for arXiv:1907.11588. Web searches for exact phrases around `characteristically dominated`, ℓ_q, L^q, and `aggregate domination` did not locate an exact existing result. The nearby literature check identified the source paper [1], Yaroslavtsev's UMD BDG theorem [2], and the Dirksen–Yaroslavtsev L^q Burkholder-Rosenthal paper [3] as surrounding context, but not as an already-recorded answer to this exact subcase.

Human Review Recommendation

Review the pushforward-of-characteristics step and the dependence on the companion scalar/Hilbert packet first. If both are accepted, the summation argument is short and robust. The most plausible next strengthening is a tail comparison that controls $\mathbb{E}\|([N^k]_\infty^{1/2})_k\|_{\ell_q}^p$ for $p \ne q$.

References

[1] I. S. Yaroslavtsev, *Local characteristics and tangency of vector-valued martingales*, arXiv:1907.11588.

[2] I. S. Yaroslavtsev, *Burkholder-Davis-Gundy inequalities in UMD Banach spaces*, arXiv:1807.05573.

[3] S. Dirksen and I. S. Yaroslavtsev, *L^q-valued Burkholder-Rosenthal inequalities and sharp estimates for stochastic integrals*, arXiv:1707.00109.

[4] Agent `agent_lane_19`, *Hilbert-valued characteristic domination for all finite p*, run packet `1907.11588_hilbert_all_p_general_characteristic_domination`, 2026.